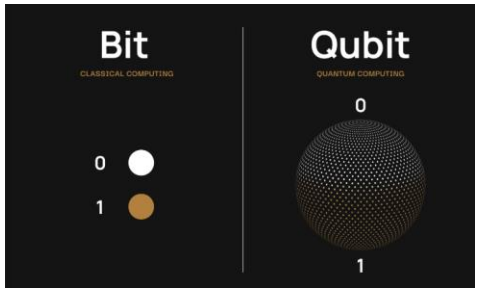
ADAPTIVE OPTIMIZATION FOR CAT QUBIT STABILIZATION

BY: REESE QUBITS

WHAT WE SAW

A critical hardware bottleneck: the scarcity of fault-tolerant architectures.

Cat qubits offer a promising solution



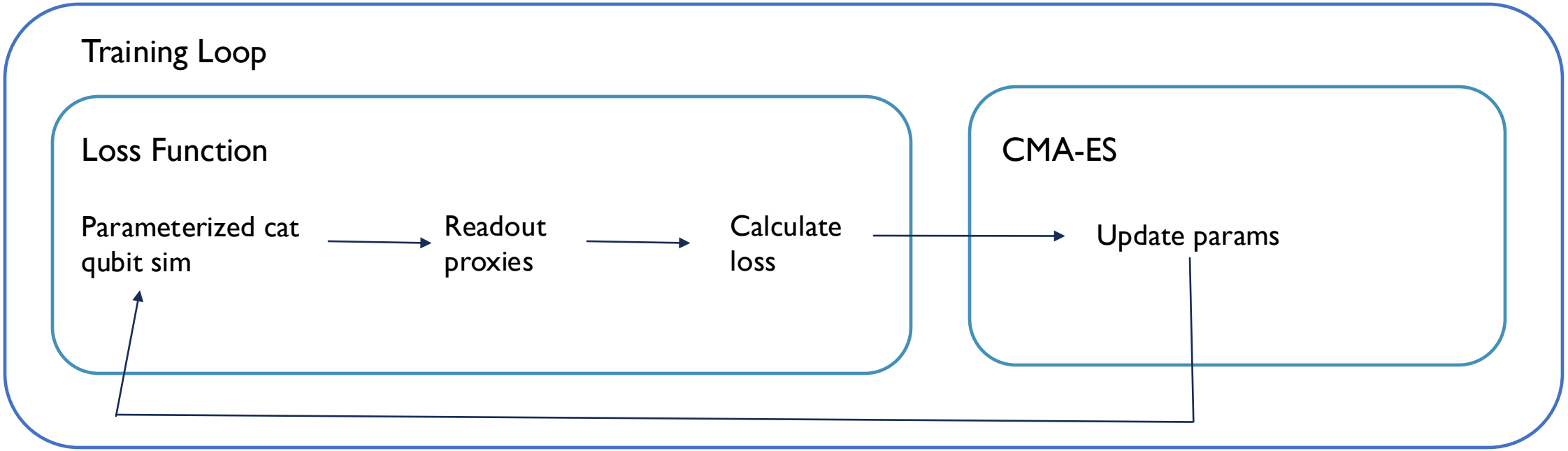
WHAT WE DID

DEVELOPED REESE QUBITS, A SYSTEM THAT CONTINUOUSLY OPTIMIZES CAT QUBIT PARAMETERS UNDER NONIDEAL CONDITIONS.

REESE QUBITS IS AN ONLINE OPTIMIZATION FRAMEWORK THAT

- Tunes control parameters
- Tracks system performance using logical proxies
- Dynamically adapts to drifts

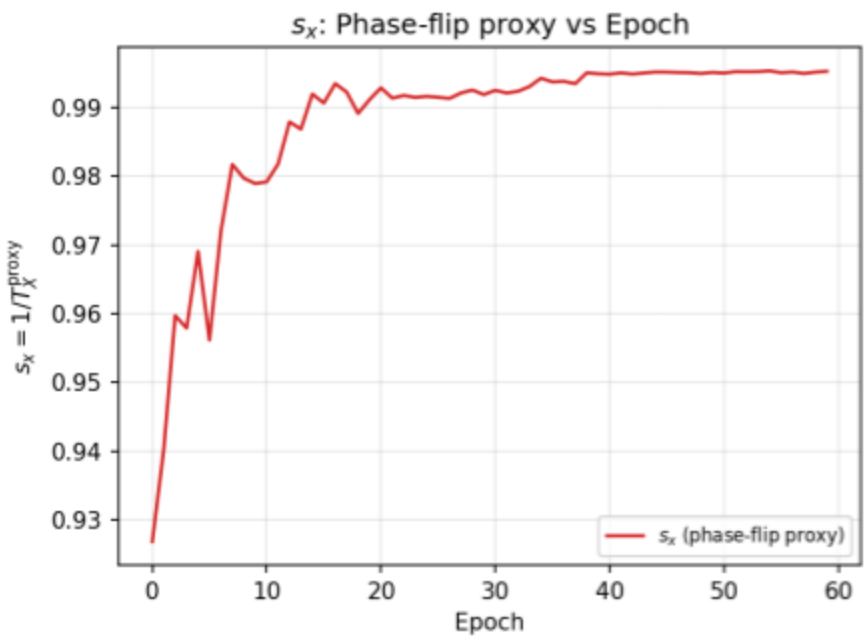
How we built it?



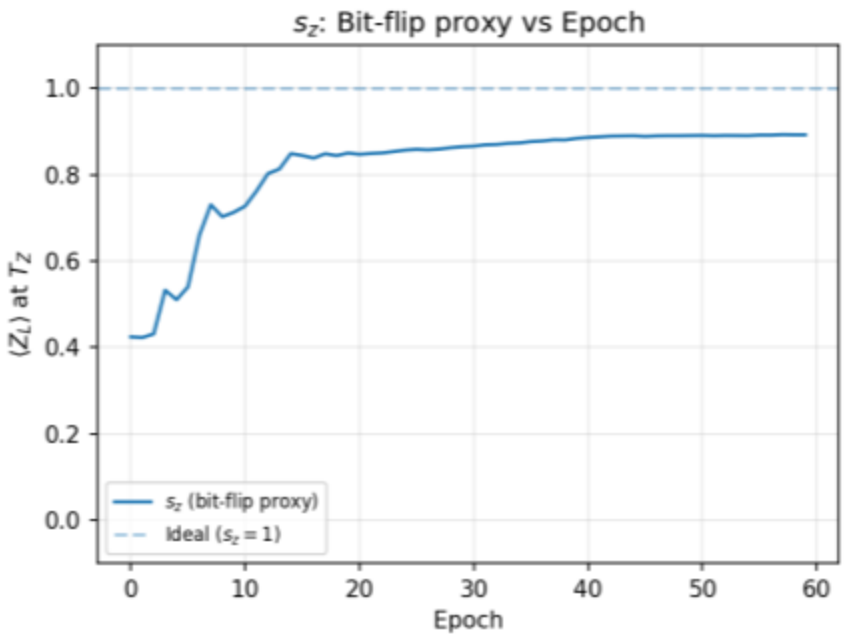
SYSTEM ARCHITECTURE

KEY DEVELOPMENT (1)

Phase-flip proxy

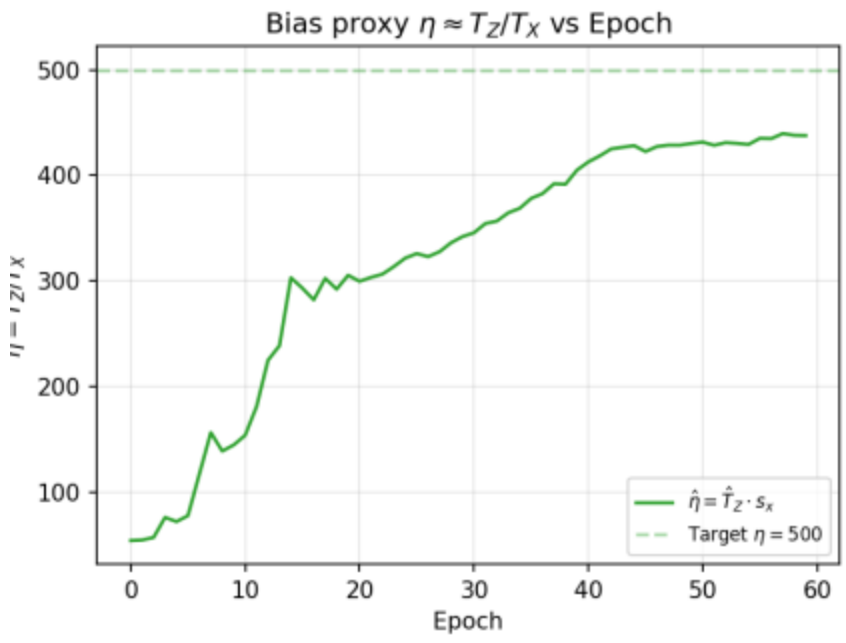


Bit-flip proxy

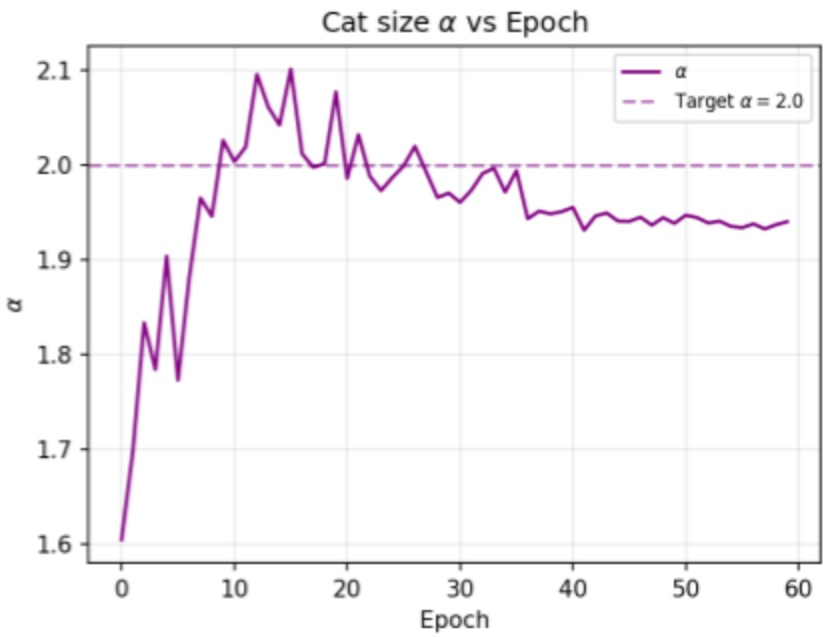


KEY RESULT (II)

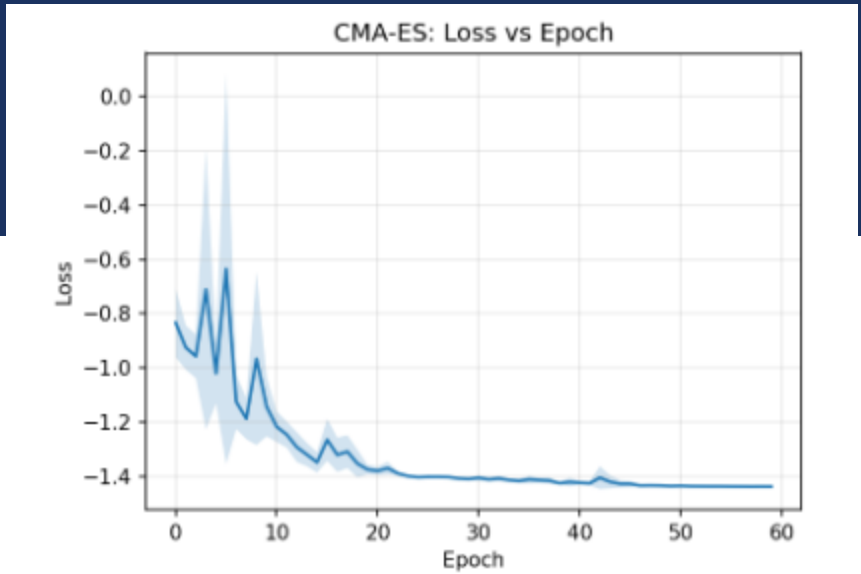
Logical bias stabilized near target



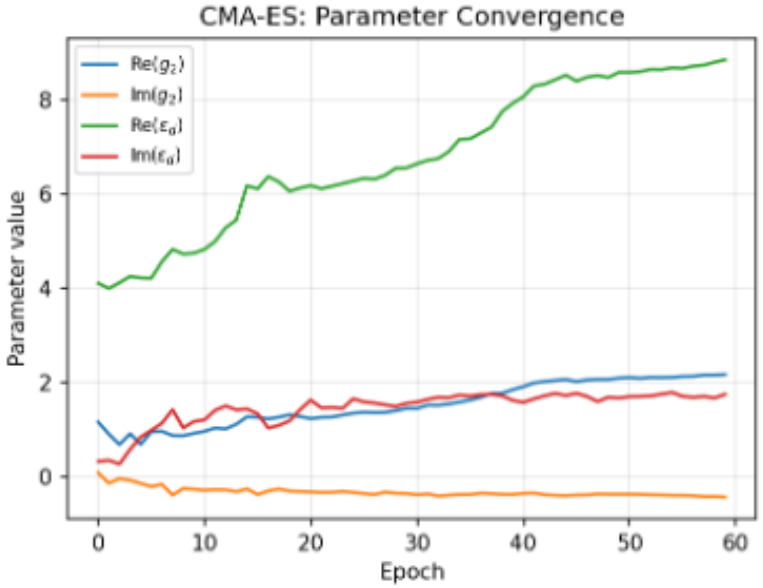
Cat size stabilized near target



LEARNING RESULTS



- Demonstrated robust convergence:
  - Loss reduced: `-0.8 → -1.37`
- Achieved strong improvement in:
  - bit-flip protection (`s_z ↑`)
  - coherence (`s_x ↑`)
- Demonstrated benefit of control knobs:
  - Detuning compensation improved recovery speed



- **Re(g_2) (blue)** → converging to **~2.1**
- **Im(g_2) (orange)** → converging to **~−0.5**
- **Re(ε_d) (green)** → converging to **~9**
- **Im(ε_d) (red)** → converging to **~1.7**

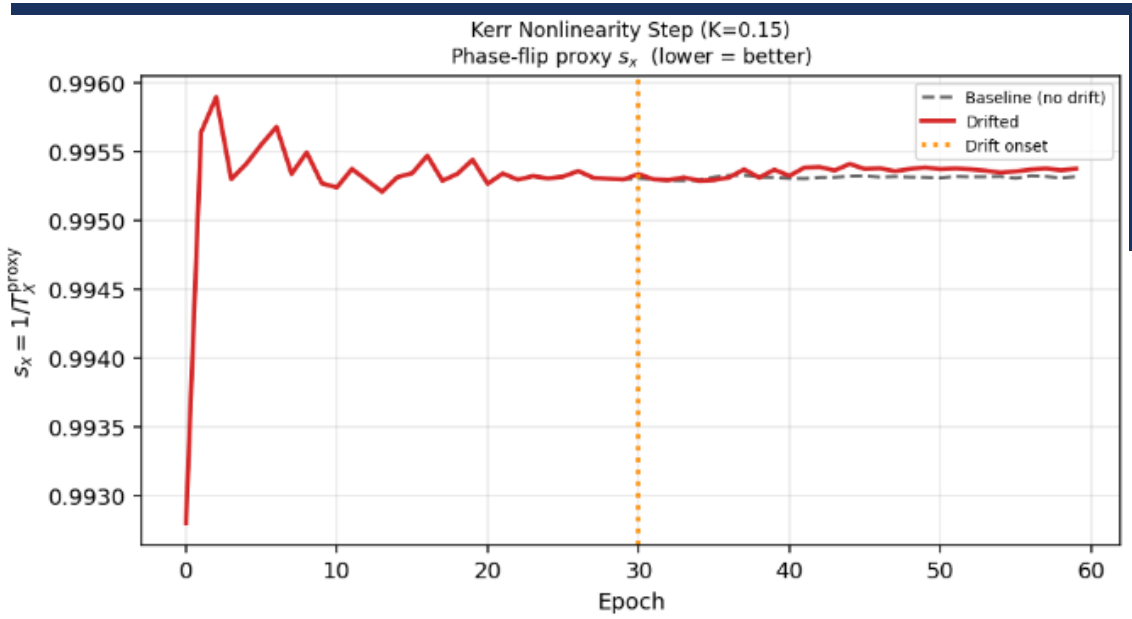
LEARNING RESULT (2)

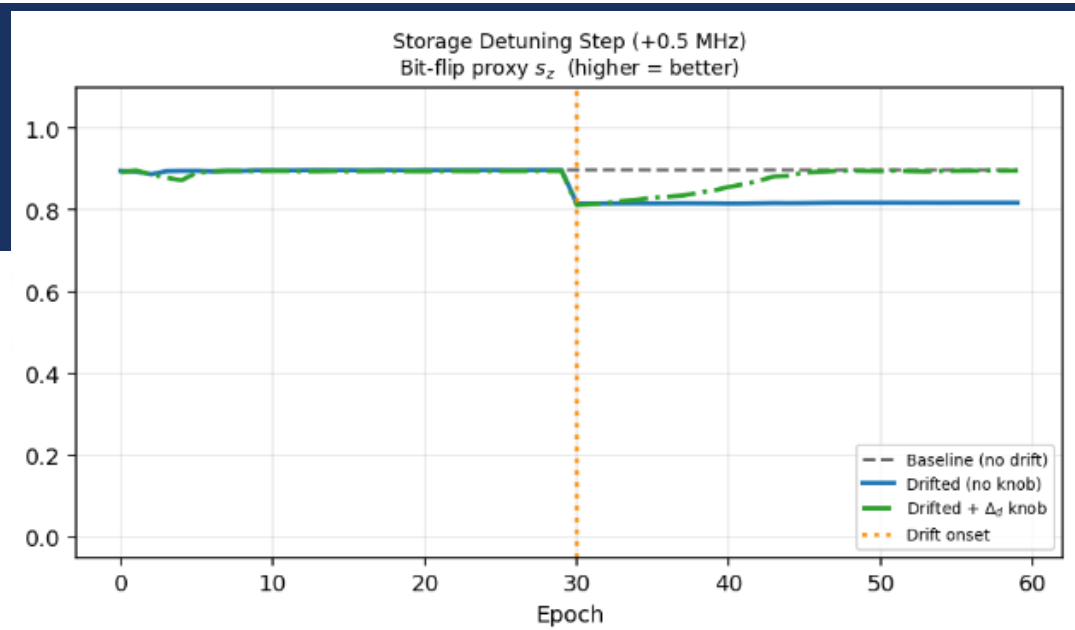
- Most disruptive:
  - g_2 amplitude drift (+30%)
- Moderate:
  - detuning drift (`Δ_a = +0.5 MHz`)
- Minimal:
  - Kerr drift (`K = 0.15 MHz`)

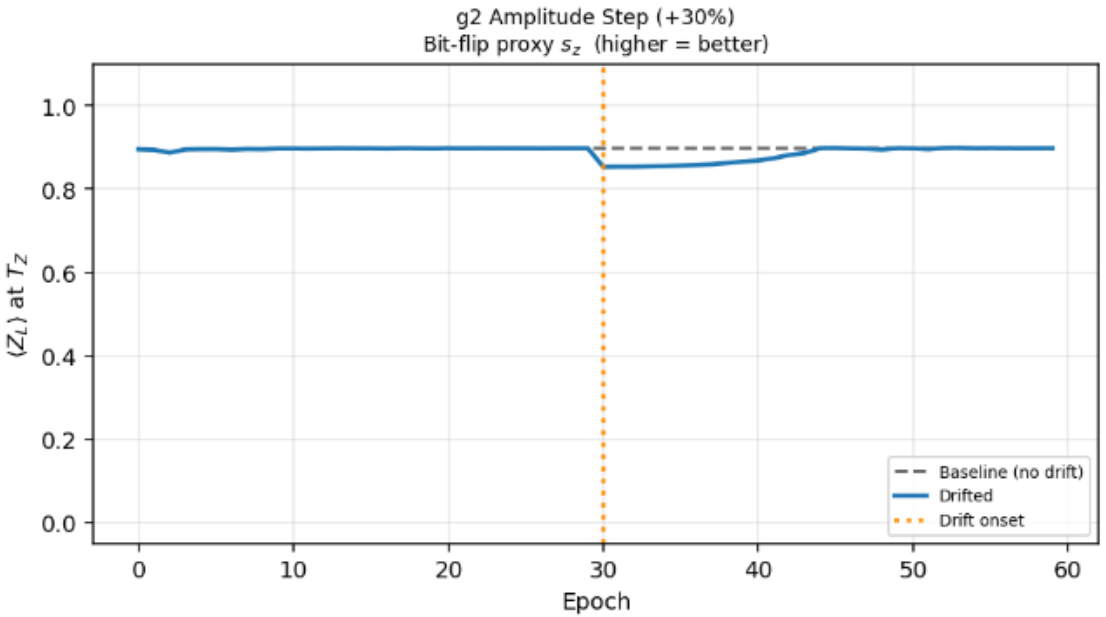
$$d = \begin{bmatrix} \delta_A & \phi & \Delta_a & K \end{bmatrix}^{\mathsf{T}},$$

- δ_A is a fractional amplitude shift of g_2,
- ϕ is an additional phase shift on g_2,
- Δ_a is a storage-mode detuning,
- K is a storage Kerr coefficient.

$$H_{\text{drift}} = (g_2^{\text{hw}})^* a^2 b^\dagger + g_2^{\text{hw}} a^{\dagger 2} b - \epsilon_d b^\dagger - \epsilon_d^* b + \Delta_a n_a + \frac{K}{2}(n_a^2 - n_a).$$


Kerr Nonlinearity Step (K=0.15)
Phase-flip proxy s_x (lower = better)
$s_X = 1/T_X^{proxy}$
Epoch
Baseline (no drift)
Drifted
Drift onset


Storage Detuning Step (+0.5 MHz)
Bit-flip proxy s_z (higher = better)
Epoch
Baseline (no drift)
Drifted (no knob)
Drifted + Δ_d knob
Drift onset


g2 Amplitude Step (+30%)
Bit-flip proxy s_z (higher = better)
$\langle Z_L \rangle$ at T_Z
Epoch
Baseline (no drift)
Drifted
Drift onset

KEY TAKEAWAY

Adaptive optimization can stabilize quantum systems under realistic drift, but achieving full fault tolerance requires better objective design, richer control knobs, and tighter integration between physics and learning.

THANK YOU!